Multiple-choice questions

|!|BCEMB0070000001|!| ★

Which of the following correctly shows the route taken by the bile pigments produced by the liver to pass out of the body with faeces?

A. liver → duodenum → gall bladder → bile duct → ileum → large intestine → anus
B. liver → gall bladder → bile duct → duodenum → ileum → large intestine → anus
C. liver → bile duct → gall bladder → pancreas → ileum → large intestine → anus
D. liver → bile duct → gall bladder → pancreas → duodenum → large intestine → anus

□

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0070000002|!|

In the human digestive system, the enzyme lactase is produced in the

A. liver.
B. pancreas.
C. stomach.
D. small intestine.

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000003|!|

Which of the following combinations about the enzyme amylase is correct?

	Site of production	*Substrate*	*Product*
A.	pancreas	starch	maltose
B.	gall bladder	starch	maltose
C.	pancreas	starch	glucose
D.	small intestine	maltose	glucose

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000004|!|

The digestion of food starts at the __X__ and ends at the __Y__.

Which of the following correctly identifies X and Y?

	X	***Y***
A.	mouth	large intestine
B.	mouth	small intestine
C.	stomach	large intestine
D.	stomach	small intestine

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0070000005|!| ★

Which of the following parts produces digestive juice which does ***not*** contain digestive enzymes?

A. mouth
B. stomach
C. liver
D. pancreas

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0070000006|!| ★

After eating a boiled egg, physical digestion begins in the

A. mouth cavity.
B. oesophagus.
C. stomach.
D. duodenum.

##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0070000007|!| ★

In which of the following parts does the emulsification of lipids take place?

A. stomach
B. liver
C. gall bladder
D. small intestine

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0070000008|!| ★

The table below shows the protein, fat and carbohydrate content in two types of food items P and Q.

Food item	Protein (g)	Fat (g)	Carbohydrate (g)
P	0.1	0.2	10.5
Q	8.2	0.0	0.1

Which of the following are the main products of complete digestion of P and Q?

	P	***Q***
A.	fatty acids	amino acids
B.	glucose	amino acids
C.	amino acids	glucose
D.	glucose	fatty acids

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0070000009|!| ★

The diagram below shows a tooth taken from an adult.



Which of the following combinations about the type of tooth shown and explanation is correct?

	Type of tooth	***Reason***
A.	premolar	it is absent in the milk teeth
B.	premolar	it has three roots
C.	molar	it has cusps
D.	molar	it has three roots

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000010|!| ★

Which of the following statements about the carbohydrases in the alimentary canal is correct?

A. They work best in both alkaline and acidic media.
B. They catalyse the breakdown of disaccharides to monosaccharides.
C. They catalyse the breakdown of polypeptides to glucose.
D. Their optimum temperature is 37 ^{o}C.

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0070000012|!| ★

Which of the following substances pass through the epithelial cells of the small intestine by both diffusion and active transport?

(1) vitamin A
(2) amino acid
(3) fatty acid

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0070000014|!| ★

The absorption of water in the small intestine is facilitated by

(1) the presence of numerous finger-like projections on its inner surface.
(2) the presence of numerous lacteals.
(3) the absorption of digested food.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0070000016|!| ★

Which of the following about the action of digestive juice is ***incorrect***?

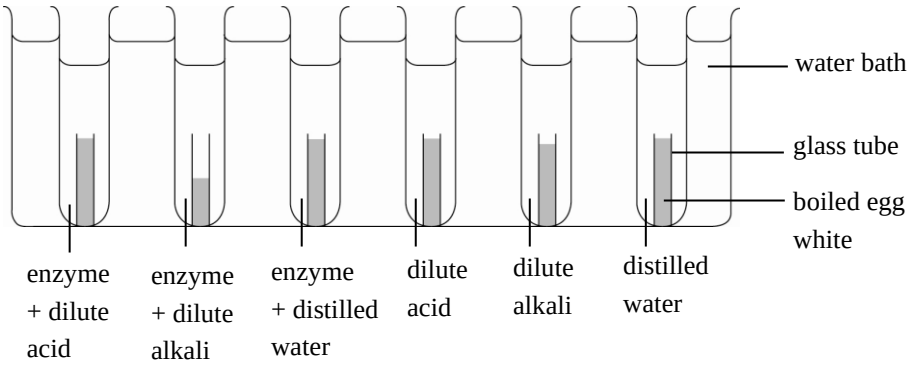
	Digestive juice	***Digestive action***
A.	bile	digestion of lipid into fatty acids and glycerol
B.	saliva	digestion of starch into maltose
C.	gastric juice	digestion of protein into peptides
D.	pancreatic juice	digestion of peptides into amino acids

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0070000017|!| ★★ SI

Peter carried out an experiment to investigate the effect of an enzyme on boiled egg white. In the experiment, he prepared six small glass tubes of boiled egg white. He then put the glass tubes into test tubes with different solutions at a constant temperature of 37 °C. The following diagram shows the results after four hours.



From which part of the digestive system is the enzyme secreted?

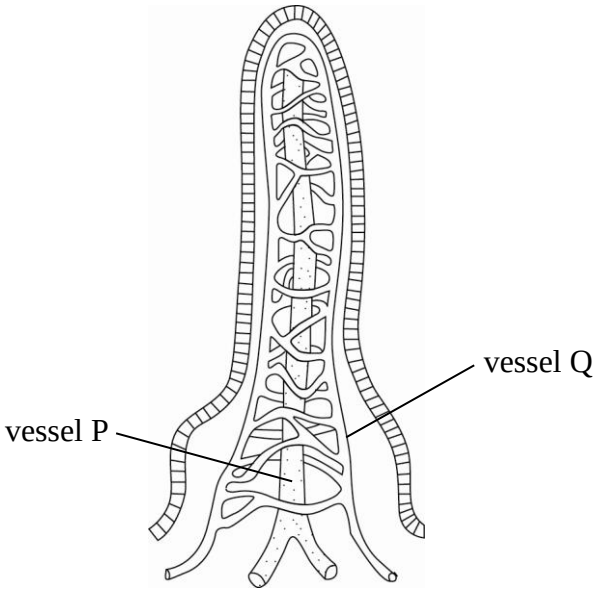
A. liver
B. pancreas
C. gastric glands
D. salivary glands

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000018|!| ★

The following diagram shows the section of an intestinal villus.



Which of the following correctly states the food substances that are absorbed mainly into vessels P and Q respectively?

	Vessel P	***Vessel Q***
A.	vitamin D	amino acids
B.	lipids	vitamin D
C.	amino acids	glucose
D.	glucose	lipids

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000019|!|

Which of the following statements about peristalsis is ***incorrect***?

A. It is a wave of muscle contraction along the alimentary canal.
B. It is under voluntary control.
C. It is stimulated by ingested food with high content of dietary fibre.
D. It can help the physical digestion of food.

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000025|!|

Pancreatic juice is released through the pancreatic duct into the

A. ileum.
B. oesophagus.
C. duodenum.
D. colon.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000026|!|

Which of the following molecules are absorbed into the blood capillaries in the villi?

(1) glucose
(2) vitamin A
(3) amino acid

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000027|!|

Which of the following is ***not*** a function of the human liver?

A. storing glycogen
B. producing bile for the emulsification of lipid
C. breaking down old red blood cells
D. storing toxic substances

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0070000028|!| ★

A man is suffered from a pancreatic disease and his pancreatic duct is blocked.

Which of the following are the effects of such condition?

(1) The chyme in the duodenum remains acidic.
(2) The lipid content of the faeces increases.
(3) The digestion of protein in the small intestine is less effective.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0070000029|!|

Excess amino acids in the human body would be

A. stored in the liver as energy reserve.
B. transported to the kidneys and excreted in urine.
C. broken down in the liver, producing urea.
D. broken down in the liver, producing bile pigment.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000030|!|

Which of the following is/are the metabolic waste(s) produced by the liver?

(1) carbon dioxide
(2) bile salt
(3) urea

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000031|!|

Which of the following are the functions of hydrochloric acid in the gastric juice?

(1) killing bacteria in food
(2) hydrolysing starch to maltose
(3) providing an acidic medium for pepsin to work

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000032|!|

Which of the following substances is absorbed into the lacteals of the villi?

A. glycerol
B. amino acid
C. vitamin C
D. triglyceride

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000033|!|

Which of the following substances is present in the intestinal juice?

A. lipase
B. bile pigment
C. protease
D. sodium hydrogencarbonate

☐

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000034|!|

Enzymes in the pancreatic juice can catalyse the breakdown of

(1) protein
(2) lipid
(3) starch

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000036|!|

Which of the following is the dental formula of a human adult?

A. $\mathrm{i}\frac{2}{3}\ \mathrm{c}\frac{1}{1}\ \mathrm{pm}\frac{2}{2}\ \mathrm{m}\frac{2}{2}$
B. $\mathrm{i}\frac{3}{2}\ \mathrm{c}\frac{1}{1}\ \mathrm{pm}\frac{2}{2}\ \mathrm{m}\frac{3}{3}$
C. $\mathrm{i}\frac{2}{2}\ \mathrm{c}\frac{1}{1}\ \mathrm{pm}\frac{2}{2}\ \mathrm{m}\frac{3}{3}$
D. $\mathrm{i}\frac{2}{3}\ \mathrm{c}\frac{1}{1}\ \mathrm{pm}\frac{2}{2}\ \mathrm{m}\frac{3}{2}$

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000038|!|

Which of the following is the major route of the transport of vitamin D to the heart after absorption?

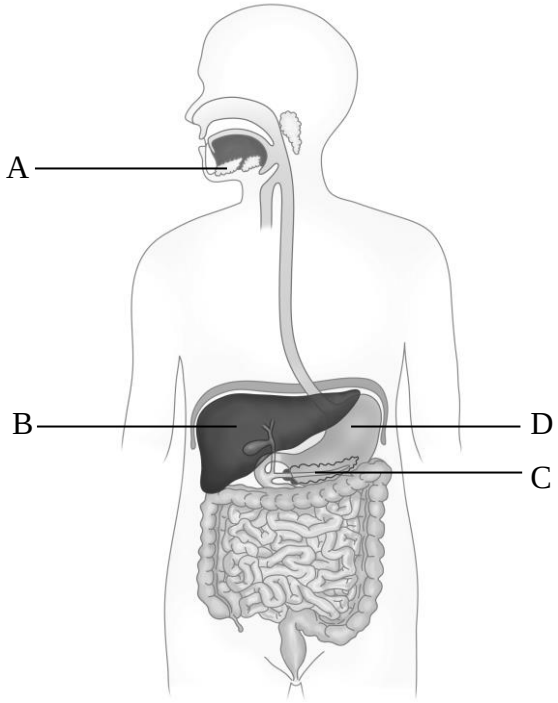
A. lacteal of villus → lymph vessels → liver → vena cava → heart
B. lacteal of villus → lymph vessels → vena cava → heart
C. blood capillaries of villus → hepatic vein → liver → vena cava → heart
D. blood capillaries of villus → hepatic portal vein → liver → vena cava → heart

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000039|!|

The diagram below shows part of the human alimentary canal.



Amylase is produced in

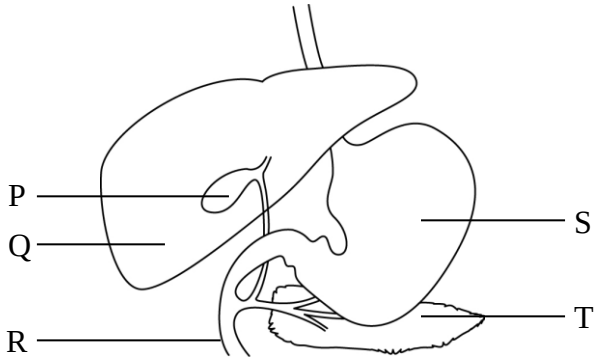
A. A only.
B. A and B only.
C. A and C only.
D. C and D only.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000041|!| ★

The diagram below shows part of the human alimentary canal and some of the associated structures.



Which structures produce enzymes for the digestion of proteins?

A. P, R and S only
B. P, R and T only
C. Q, S and T only
D. R, S and T only

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000042!| ★

Which of the following parts of the alimentary canal does ***not*** produce any digestive enzymes?

A. stomach
B. liver
C. pancreas
D. duodenum

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000043|!|

Which of the following structures produces both digestive enzymes and hormones?

A. gall bladder
B. pancreas
C. duodenum
D. ileum

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000044|!|

Which of the following will happen if the bile duct is obstructed?

A. No more pancreatic enzyme is made.
B. The pancreatic juice cannot enter the duodenum.
C. Lipid cannot be digested.
D. The lipid content in the faeces increases.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000046|!|

Which of the following functions is ***not*** performed by the human stomach?

A. secretion of lipase to digest lipid
B. secretion of protease to digest proteins
C. churning of food
D. absorption of water

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000047|!|

Which of the following is ***not*** a function of the liver?

A. production of bile
B. storage of glycogen
C. deamination of excess amino acids
D. removal of urea from the blood

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000049|!|

In which of the following parts of the alimentary canal does peristalsis take place?

(1) oesophagus
(2) duodenum
(3) colon

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000050|!|

Which of the following digestive juices contain(s) enzymes?

(1) bile
(2) saliva
(3) intestinal juice

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000051|!|

The following shows the fate of the proteins in food eaten by a human.

proteins in food
↓ (1)
proteins in the mouth
↓ (2)
amino acids in the small intestine
↓ (3)
amino acids in the blood
↓ (4)
amino acids taken up and used by body cells

Which of the following combinations about processes (1) to (4) is correct?

	(1)	*(2)*	*(3)*	*(4)*
A.	egestion	ingestion	absorption	assimilation
B.	egestion	digestion	assimilation	absorption
C.	ingestion	digestion	assimilation	absorption
D.	ingestion	digestion	absorption	assimilation

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000052|!|

Which of the following statements about the emulsification of oil by bile is correct?

A. It takes place in the gall bladder.
B. It increases the surface area of oil.
C. This provides an optimum pH for lipase to act on oil.
D. The rate of emulsification is the highest at 37 ^{o}C.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000054|!|

Which of the following parts of the alimentary canal absorbs most of the water in the diet?

A. colon
B. small intestine
C. stomach
D. rectum

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000055|!|

Which of the following substances is carried by the lymph after absorption?

A. vitamin A
B. vitamin C
C. amino acid
D. calcium

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000056|!|

Which of the following digestive juices contain(s) sodium hydrogencarbonate?

(1) bile
(2) intestinal juice
(3) pancreatic juice

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0070000057|!|

Which of the following are the adaptive features of the ileum for absorbing nutrients?

(1) presence of microvilli
(2) each villus contains a dense network of blood capillaries and a central lacteal
(3) presence of digestive enzymes on its epithelial cells

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000058|!|

Which of the following may be the consequence if a great length of the large intestine of a patient is removed?

A. Dietary fibre cannot be digested.
B. The faeces may become more watery.
C. The patient may become thin as fewer nutrients are absorbed.
D. The patient may be deficient of some minerals as minerals are absorbed in the large intestine.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000059|!|

Which of the following about the hepatic portal vein is correct?

A. It delivers fatty acid from the liver to the heart.
B. It delivers glucose from the small intestine to the liver.
C. It delivers lipid from the small intestine to the liver.
D. It delivers amino acid from the liver to the heart.

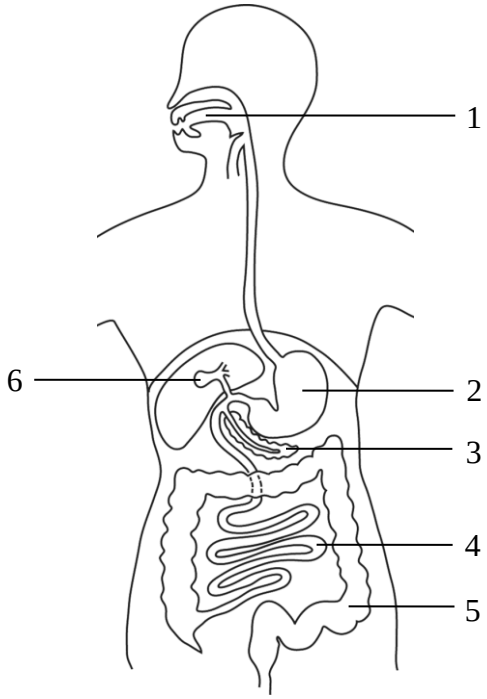
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0070000060|!| ★

***Directions*:** The following two questions refer to the diagram below, which shows the alimentary canal and some of its associated structures.



In which labelled part(s) does the absorption of water take place?

A. 4 only
B. 5 only
C. 4 and 5 only
D. 2, 4 and 5

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0070000061|!| ★

Which labelled structures produce carbohydrases?

A. 1 and 3 only
B. 1 and 6 only
C. 3 and 4 only
D. 1, 3 and 4 only

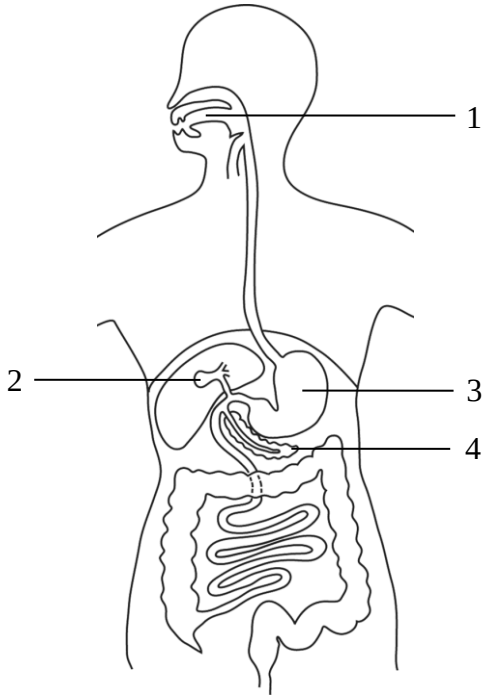
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0070000062|!| ★★

Directions: The following two questions refer to the diagram below, which shows the alimentary canal and some of its associated structures.



Digestive juice was extracted from structure 2 and then added to a test tube containing oil. The pH of the mixture increased. This is because the digestive juice from structure 2

A. contains bile pigment.
B. contains bile salts which emulsify the oil.
C. contains enzymes which catalyse the breakdown of lipids to glycerol and fatty acids.
D. contains alkaline salts.

□

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0070000063|!| ★★

Which structures are involved in the physical and chemical digestion of egg white?

	Physical digestion of egg white	*Chemical digestion of egg white*
A.	3 only	3 only
B.	3 only	3 and 4
C.	1 and 3	3 only
D.	1 and 3	3 and 4

□

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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0070000064|!| ★★ SI

Directions: The following two questions refer to the following investigation. Digestive juice X was extracted from a healthy man. It was then mixed with egg-white solution and pH buffers of different pH values. The table below shows the preparation of the four test tubes used and the results after one hour.

Test tube	Content	Results
P	1 cm^3 of juice X, 1 cm^3 of egg white solution and 1 cm^3 of buffer solution of pH 2	Clear solution
Q	1 cm^3 of juice X, 1 cm^3 of egg white solution and 1 cm^3 of buffer solution of pH 7	Cloudy solution
R	1 cm^3 of juice X, 1 cm^3 of egg white solution and 1 cm^3 of buffer solution of pH 10	Cloudy solution
S	1 cm^3 of boiled juice X, 1 cm^3 of egg white solution and 1 cm^3 of buffer solution of pH 2	Cloudy solution

What conclusion can be drawn based on the results?

A. Egg white can be broken down by an acidic solution.
B. Juice X works best at pH 10.
C. Juice X does not contain enzymes.
D. Juice X contains a heat-sensitive substance which breaks down egg white.

□

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0070000065|!| ★★ SI

Digestive juice X was most likely extracted from

A. the mouth cavity.
B. the stomach.
C. the gall bladder.
D. the duodenum.

□

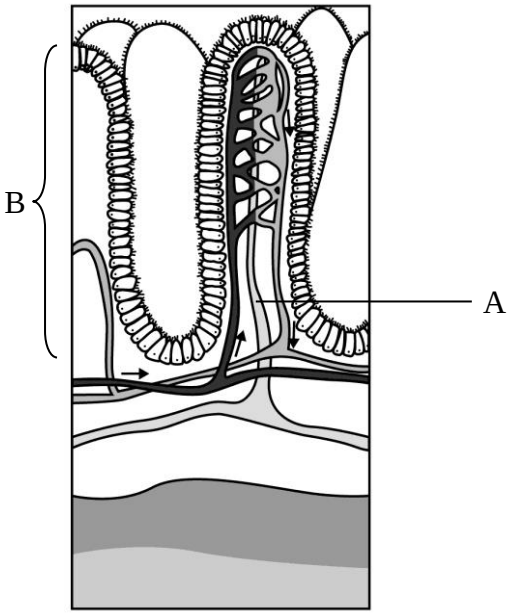
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##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0070000066|!|

Directions: The following two questions refer to the diagram below which shows the longitudinal section of the ileum.



Which of the following substances is absorbed into A?

A. dietary fibre
B. lipid
C. glucose
D. protein

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000067|!|

Which of the following are the adaptations of structure B for the absorption of food?

(1) The epithelial cells of structure B have microvilli.
(2) The epithelial cells of structure B have enzymes on the cell membranes.
(3) There is a dense network of capillaries within structure B.

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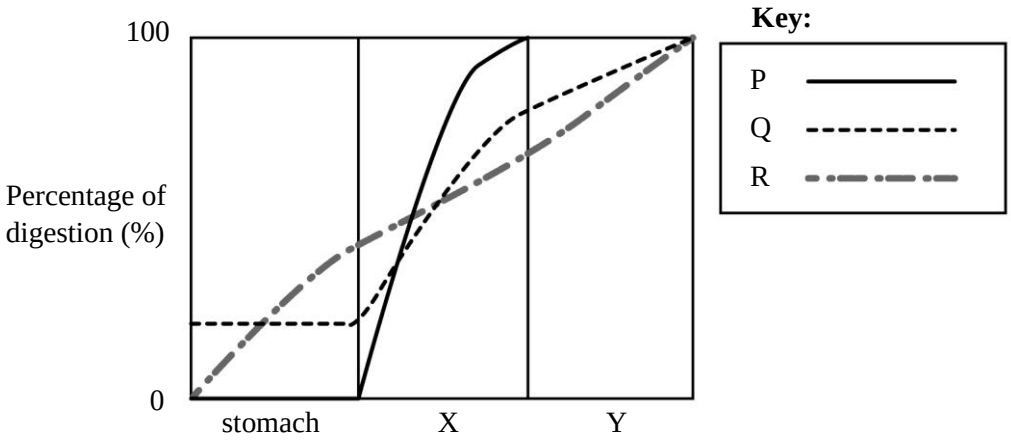
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0070000068|!| ★

Directions: The following two questions refer to the graph below which shows the percentage of digestion of different food substances along different regions of the human alimentary canal (stomach, structures X and Y). Curves P, Q and R represent three different food substances, which can be carbohydrates, lipids or proteins.



Which of the following combinations correctly identifies substances P and R?

	P	***R***
A.	lipid	carbohydrate
B.	protein	carbohydrate
C.	lipid	protein
D.	carbohydrate	protein

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000069|!| ★

Which of the following combinations correctly identifies structures X and Y?

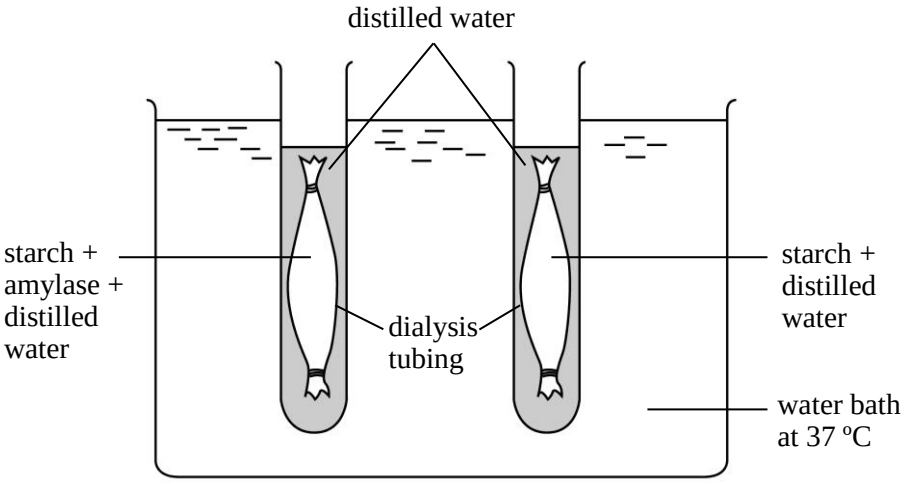
	X	***Y***
A.	small intestine	large intestine
B.	duodenum	ileum
C.	duodenum	colon
D.	ileum	duodenum

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000070|!| ★ SI

Directions: The following two questions refer to the diagram below which shows a model representing part of a digestive system.



Which of the following combinations correctly states what the components of the model are representing?

	Dialysis tubing	***Solution surrounding the dialysis tubing***
A.	small intestine	intestinal juice
B.	small intestine	blood
C.	blood vessel	blood
D.	blood vessel	intestinal juice

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000071|!| ★ SI

Which of the following should be done to ensure that the results are valid?

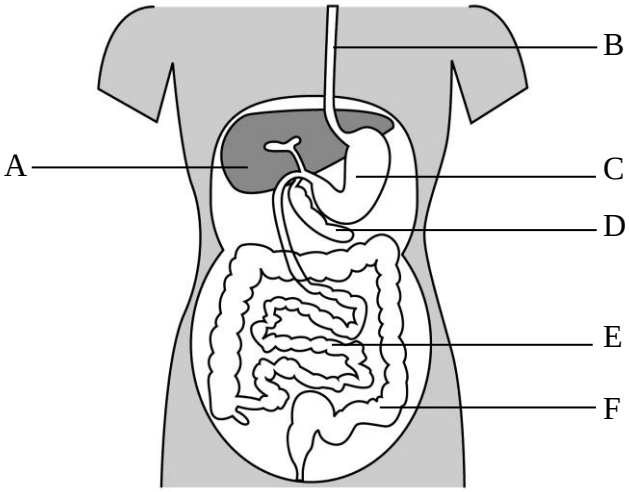
A. rinse the dialysis tubing with distilled water before they are put into the test tubes
B. sterilize the dialysis tubing before use
C. test the water outside the dialysis tubing for starch
D. rinse the outside surface of the test tubes with distilled water before they are put into the water bath

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0070000072|!|

Directions: The following two questions refer to the diagram below which shows some of the structures concerned with digestion.



Which of the labelled structures produces enzymes for the digestion of lipids?

A. organ A
B. organ C
C. organ D
D. organ E

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000073|!|

Which of the labelled structures secretes hormones to regulate blood glucose level?

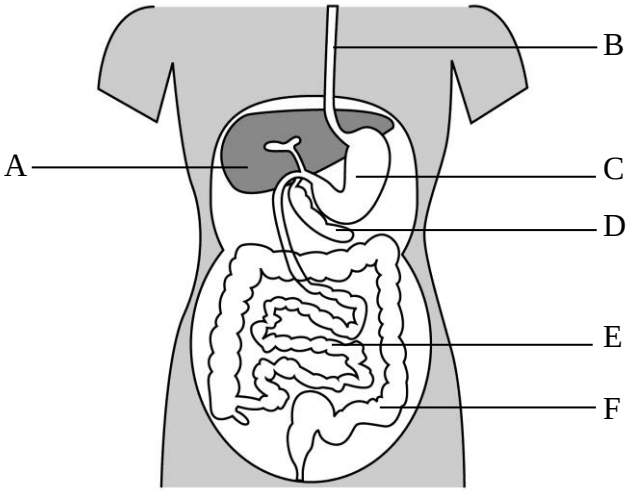
A. organ A
B. organ C
C. organ D
D. organ E

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000074|!|

Directions: The following two questions refer to the diagram below which shows some of the structures concerned with digestion.



Churning of food occurs in

A. organ B.
B. organ C.
C. organ E.
D. organ F.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000075|!|

Most of the water in food is absorbed in

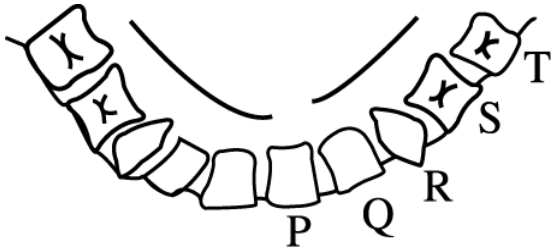
A. organ B.
B. organ C.
C. organ E.
D. organ F.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000076|!| ★

Directions: The following two questions refer to the diagram below which shows the milk teeth in the lower jaw of a three-year-old child.



Which of the following combinations correctly identities the types of the teeth shown?

	P	***Q***	***R***	***S***	***T***
A.	incisor	canine	canine	premolar	molar
B.	incisor	canine	premolar	molar	molar
C.	incisor	incisor	canine	premolar	molar
D.	incisor	incisor	canine	molar	molar

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0070000077|!| ★

Which of the following teeth is used for biting and cutting food?

A. Q
B. R
C. S
D. T

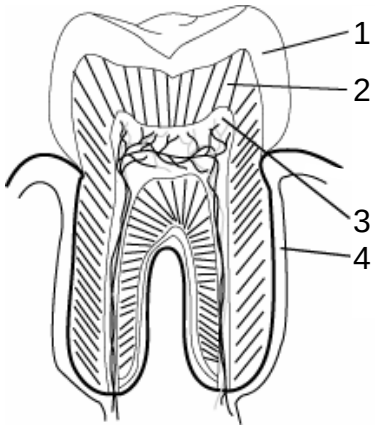
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##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000078|!|

Directions: The following two questions refer to the diagram below which shows a human tooth.



The hardest part of the tooth is

A. 1.

B. 2.

C. 3.

D. 4.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000079|!|

Given that the dental formula of a three-year old child is $\frac{212}{212}$, how many teeth of the type shown in the diagram does the child have?

A. 2

B. 4

C. 8

D. 12

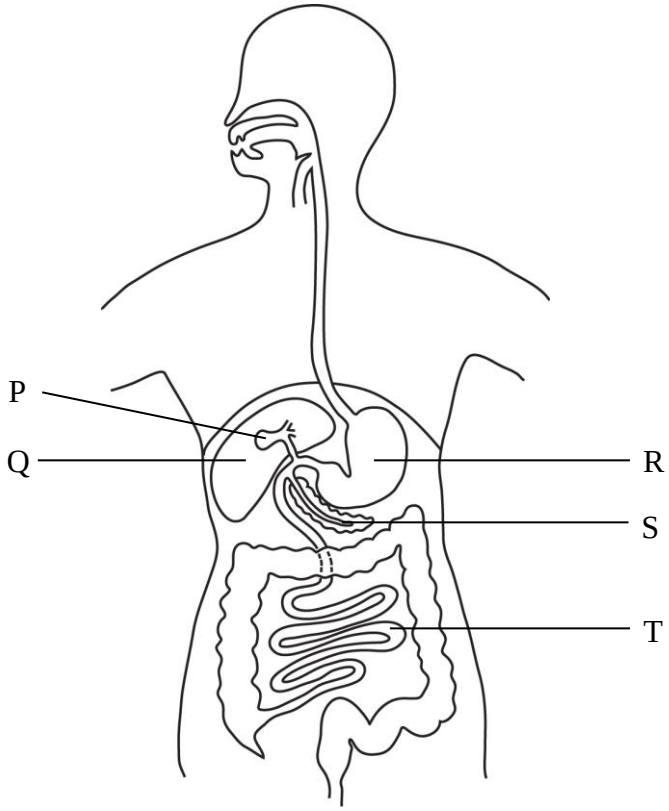
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0070000080|!|

Directions: The following two questions refer to the diagram below which shows the human alimentary canal and the associated glands.



Which of the following structures secretes a fluid with the lowest pH?

A. structure P
B. structure Q
C. structure R
D. structure S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000081|!|

Which of the following structures produces bile?

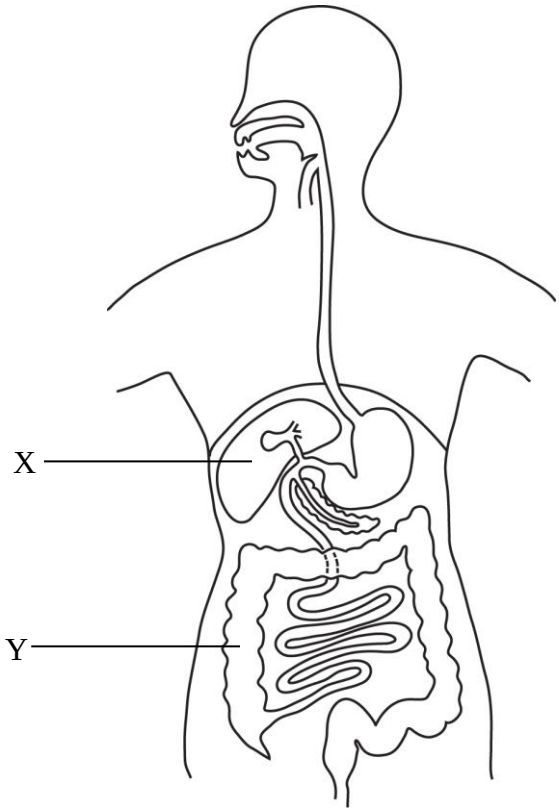
A. structure P
B. structure Q
C. structure S
D. structure T

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000082|!|

Directions: The following two questions refer to the diagram below which shows the human alimentary canal and the associated structures.



Which of the following functions is carried out by structure Y?

A. absorption of water
B. absorption of amino acids
C. storage of undigested food
D. formation of faeces

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000083|!|

Blood is carried to structure X through the

(1) hepatic portal vein.
(2) hepatic vein.
(3) hepatic artery.

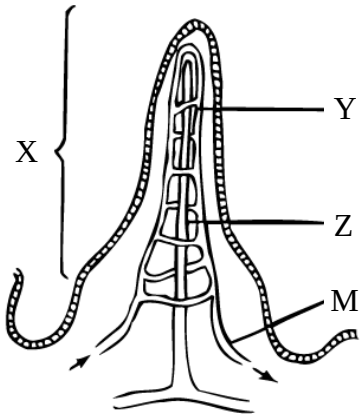
A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000084|!|

Directions: The following three questions refer to the diagram below which shows the internal structure of a certain part of the human alimentary canal.



Which of the following combinations correctly identifies X, Y and Z?

	X	***Y***	***Z***
A.	microvillus	blood capillary	lacteal
B.	villus	blood capillary	lacteal
C.	villus	lacteal	blood capillary
D.	microvillus	lacteal	blood capillary

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000085|!|

Which of the following combinations correctly shows the substances entering structures Y and Z?

	Entering Y	***Entering Z***
A.	protein	lipid
B.	fatty acid	glucose
C.	amino acid	lipid
D.	glucose	glycerol

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000086|!|

To which of the following organs does the blood in vessel M first go?

A. heart
B. lungs
C. liver
D. kidney

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0070000087|!|

Directions: The following two questions refer to the diagrams below. **Diagram X** shows the drops of lipid in the stomach and **Diagram Y** shows the oil droplets in the duodenum.

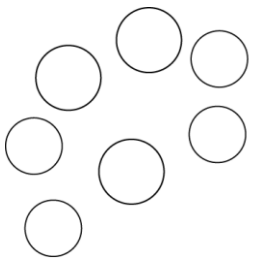


Diagram X

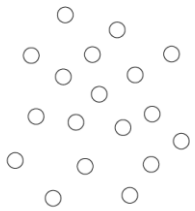


Diagram Y

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Which of the following processes leads to the formation of the oil droplets in **Diagram Y**?

A. digestion
B. emulsification
C. assimilation
D. neutralization

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0070000088|!|

What is the significance of changing the lipid drops in **Diagram X** to the oil droplets in **Diagram Y**?

A. This can break down the lipid into fatty acids and glycerol.
B. This can facilitate peristalsis.
C. This can increase the surface area for the action of lipase.
D. Smaller oil droplets can diffuse across the epithelium of the ileum.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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